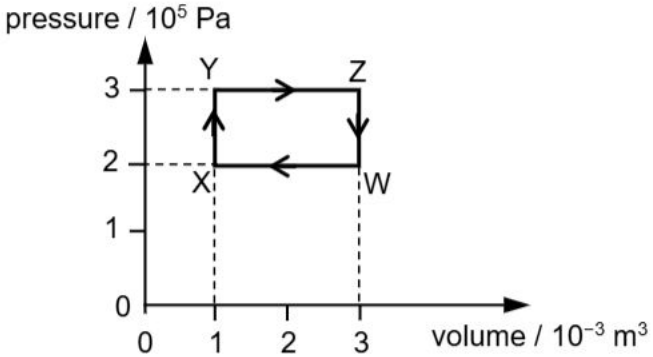
18 A fixed mass of ideal gas undergoes the cycle of changes represented by WXYZW in the diagram. There is a decrease in internal energy of 600 J from W to X and an increase in internal energy of 150 J from X to Y.



Which of the following set of information is correct for one complete cycle WXYZW?

	net increase in internal energy / J	net work done on gas / J	net heat absorbed by gas / J
A	0	200	−200
B	0	−200	200
C	450	200	250
D	−450	−200	−250

Ans: **B**

For cyclic process, $\Delta U = 0$

Since its an clockwise cycle, there is net negative work done on the gas

Net work done on gas = area enclosed

$W = -((3-2)\times 10^5)((3-1)\times 10^{-3})$

$= -200$ J

$\Delta U = Q + W$

$0 = Q + (-200)$

$Q = 200$ J